

Indian Institute of Technology Dharwad
Department of EECE
EE 601L: Neural Networks and Deep Learning Lab

LAB-4

10/02/ 2026

Analysis of Vanishing and Exploding Gradient Problems in Deep Neural Networks with Varying Depth and Activation Functions

Objective

To investigate the vanishing and exploding gradient problems in neural networks by:

1. Implementing forward and backward propagation **from scratch** for a shallow neural network to understand gradient computation using the chain rule.
2. Studying the effect of **network depth** and **activation functions** on gradient flow in deep neural networks.
3. Experimentally comparing gradient behavior using different activation functions across networks with increasing numbers of hidden layers.

Problem Statement

Train neural networks on a **non-linearly separable dataset** and analyze how gradient magnitudes change during backpropagation as the **number of hidden layers increases** and as **different activation functions** are used. Identify conditions under which gradients vanish or explode and justify the observed behavior.

Task 1: Dataset Generation and Preprocessing

1. Generate a non-linearly separable dataset using `make_moons()`.
2. Normalize the dataset and split it into training and testing sets.

Task 2: Two-Layer Neural Network (From Scratch)

1. Design a two-layer neural network (one hidden layer and one output layer).
2. Implement the **forward pass and backward pass manually**, without using any automatic differentiation tools.
3. Train the network separately using : a) Sigmoid b) tanh c) ReLU activation
4. Track and plot **gradient norms** for each activation function during training.

Task 3: Deep Neural Networks with (5+ Layers)

1. Using a deep learning framework (TensorFlow or PyTorch), implement neural networks with varying depths (5+ hidden layers).
2. For each depth, train separate models using: a) Sigmoid b) tanh c) ReLU activation
3. Keep all other hyperparameters constant.

Task 4: Gradient Monitoring and Analysis

1. Extract gradients during backpropagation.
2. Compute layer-wise gradient norms.
3. Plot gradient norms versus training epochs for:
 - Different depths
 - Different activation functions

Task 5: Comparative Study and Interpretation

1. Compare gradient behavior across shallow and deep networks.
2. Analyze how vanishing gradients worsen with increasing depth.
3. Explain why certain activation functions mitigate gradient degradation.
4. Identify scenarios where exploding gradients may occur.

Implementation Instructions

- Each task must be implemented **separately and in sequence**.
- **Do not combine code from different tasks** in a single section.
- Complete **Task 1 fully before starting Task 2**.
- Dataset generation and preprocessing must appear **only in Task 1**.
- Model implementation and training must appear **only in Task 2**.
- For Task 2, each activation function must be trained **independently**.

Submission Requirements

Code: Submit clean and well-commented Python code for all tasks.

Report: a) Include labeled plots and detailed explanations.

b) Answer the posed questions comprehensively.